

Tools for Troubleshooting Theory

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How Multimeter Helps
1. Computer SMPS not working	Measures the output voltage of the SMPS to check whether it is providing the correct power.
2. Smartphone not charging	Checks voltage and continuity in the charging port, cable, or charging circuit to identify a power problem.
3. Motherboard/PCB short circuit	Measures resistance or continuity between power lines and ground to help detect a possible short circuit.

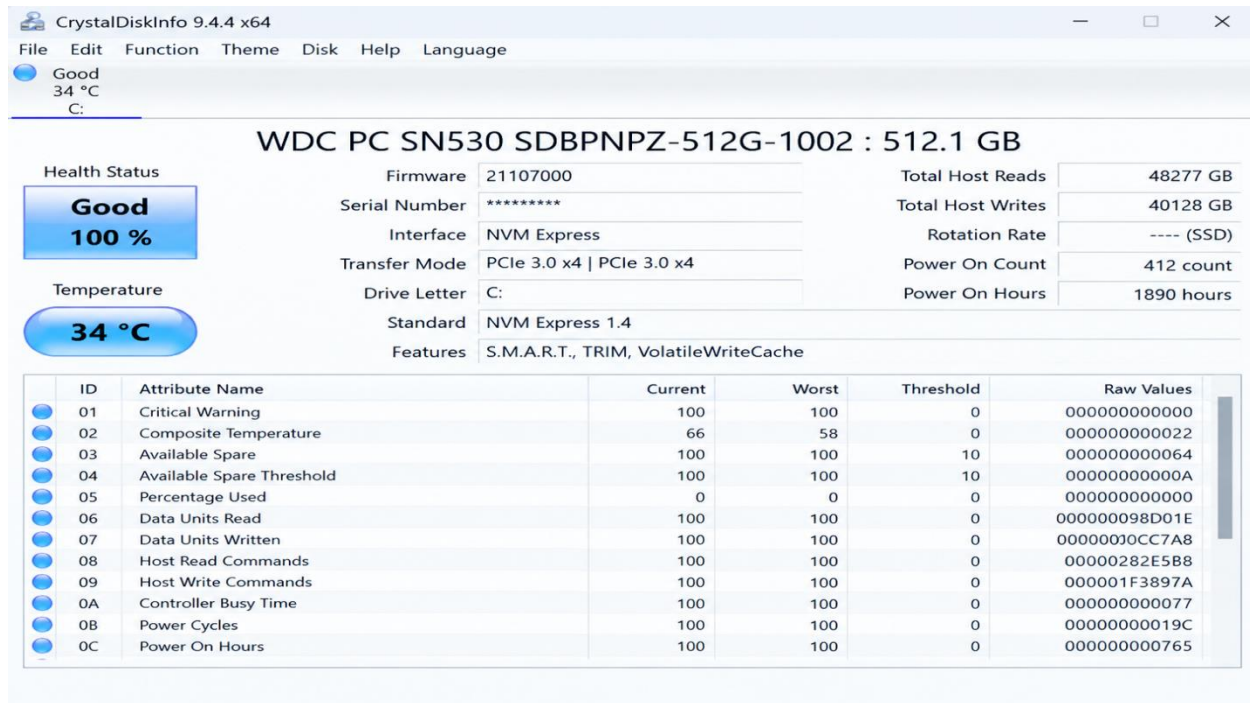
2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

- The POST card displays **hexadecimal codes** while the BIOS checks the computer's hardware during startup. In the demonstrated video, the codes change as different hardware tests are performed, and a completed POST is shown by the final code **8080**. The card also has LED indicators for power voltages and signals, helping identify problems such as a **power-supply or CPU-related fault**.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tools	What it checks	Example Problem Detected
Multi meter	Measures voltage, resistance, current, and continuity.	Faulty SMPS – detects incorrect output voltage.
POST Card	Checks hardware during startup using POST/BIOS error codes.	RAM or CPU failure – POST code indicates a hardware fault.
Software Diagnostics	Checks system health, temperature, storage, RAM, and hardware status through software.	Failing HDD/SSD – diagnostic software reports poor drive health or SMART errors.